

Convex structure of unital CP map

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1 Previous result

The following results can be found in [2].

Proposition 1. *Suppose $\Phi : \mathbf{M}_n \rightarrow \mathbf{M}_m$ is completely positive map. Given unitaries $U \in \mathbf{M}_n$ and $V \in \mathbf{M}_m$, define $\Psi : \mathbf{M}_n \rightarrow \mathbf{M}_m$ by $\Psi(X) = V^* \Phi(U^* X U) V$. Then, Ψ is also completely positive. Furthermore, Ψ is unital/trace preserving if and only if Φ is. Also, if $\Phi(X) = \sum_{j=1}^r F_j^* X F_j$, then $\Psi(X) = \sum_{j=1}^r (U F_j V)^* X (U F_j V)$.*

If two maps have $\Psi(X) = V^* \Phi(U^* X U) V$, we say Φ, Ψ are unirailly equivalent. Next, since for a positive linear map Φ , we have $\Phi(X^*) = \Phi(X)^*$. We can focus our study on hermitian matrices $\mathbf{H}_n$

Theorem 1. *Let $A \in \mathbf{H}_n$, $B \in \mathbf{H}_m$, then there exist two unitaries $U \in \mathbf{M}_n$, $V \in \mathbf{M}_m$ such that $U^* A U$ and $V^* B V$ are diagonal matrices, with $\mathbf{a}_i = (a_{i1}, a_{i2}, \dots, a_{in})$ and $\mathbf{b}_i = (b_{i1}, b_{i2}, \dots, b_{im})$. Then, the followings are equivalent.*

1. *There exists a completely positive map $\Phi : \mathbf{M}_n \rightarrow \mathbf{M}_m$ such that $\Phi(A) = B$*
2. *There is an $n \times m$ non-negative matrix $D = (d_{ij})$ such that $(b_{ij}) = (a_{ij})D$. In this case, for $1 \leq j \leq m$, let F_j be the $n \times n$ matrix having j -th column equal to $(\sqrt{d_{1j}}, \sqrt{d_{2j}}, \dots, \sqrt{d_{nj}})$ and zero elsewhere. Thus, we have*

$$B = \sum_{j=1}^m (U F_j V)^* A (U F_j V)$$

Furthermore,

1. *Φ is unital if and only if D is column stochastic.*
2. *Φ is trace preserving if and only if D is row stochastic.*
3. *Φ is unital quantum channel if and only if D is doubly stochastic.*

2 Convex structure

Lemma 1. *Let $Y \subset X$ be a convex subset of a convex set X in a vector space. Then, every extreme point of X in Y is an extreme point in Y .*

Proof. Suppose $x \in Y$ is not an extreme point of Y . Then there is point $p, q \in Y$ such that $x = \lambda p + (1 - \lambda)q$. Since $p, q \in Y \subset X$, x is not an extreme point of X , which is a contradiction. $\square$

Proposition 2. *Given $A \in \mathbf{H}_n$ and $B \in \mathbf{H}_m$ and $\mathbf{a}, \mathbf{b}$ be the corresponding eigenvalues. Let $\mathbf{CS}[n, m]$ be the set of all column stochastic matrices that is size $n \times m$ with $\mathbf{b} = \mathbf{a}D$. Then, there is an mapping from $\mathbf{CS}[n, m]$ to the extreme point of the set $\mathbf{ICP}[\mathbf{M}_n, \mathbf{M}_m, A, B] = \{\Phi \in \mathbf{CP}[\mathbf{M}_n, \mathbf{M}_m, I] : \Phi(A) = B\}$*

Proof. Let $D \in \mathbf{CS}[n, m]$. Then, by the construction from theorem 1, we have a map $\Phi(X) = \sum_{j=1}^m (UF_j V)^* X (UF_j V)$. Now, we will check Φ is an extreme point of $\mathbf{CP}[\mathbf{M}_n, \mathbf{M}_m, I]$. From [1], we need to check the set $\{(UF_i V)^* (UF_j V)\}_{ij}$ is linearly independent. Since $(UF_i V)^* (UF_j V) = V^* F_i^* F_j V$, through direct computation, we have that $F_i^* F_j = \lambda_{ij} E_{ij}$ for $\lambda_{ij} = \sum_{k=1}^n \sqrt{d_{ki}} \sqrt{d_{kj}}$ and E_{ij} be the standard basis of $\mathbf{M}_n$. Thus, Φ is the extreme point of $\mathbf{CP}[\mathbf{M}_n, \mathbf{M}_m, I]$ satisfying $\Phi(A) = B$. Next, since $\Phi \in \mathbf{ICP}[\mathbf{M}_n, \mathbf{M}_m, A, B]$, by lemma 1, we have Φ be the extreme point of $\mathbf{ICP}[\mathbf{M}_n, \mathbf{M}_m, A, B]$ $\square$

Proposition 3. *Let τ be the mapping introduced in thm 1 and prop 2. Then, τ is injective.*

Proof. Let $D_1, D_2 \in \mathbf{CS}[n, m]$ satisfying $\tau(D_1) = \tau(D_2)$. From Choi's isomorphism, we have $\tau(D_1)(E_{pq}) = \tau(D_2)(E_{pq})$ for all p, q . Then, we have the equation, $\sum_{j=1}^m F_j^* U^* E_{pq} U F_j = \sum_{j=1}^m \tilde{F}_j^* U^* E_{pq} U \tilde{F}_j$. Now, let f_j be the j th column of F_j , through computation, we have $\sum_{j=1}^m (f_j^* U^* E_{pq} U f_j) E_{jj} = \sum_{j=1}^m (\tilde{f}_j^* U^* E_{pq} U \tilde{f}_j) E_{jj}$. This means that $f_j^* U^* E_{pq} U f_j = \tilde{f}_j^* U^* E_{pq} U \tilde{f}_j$ for all p, q, j . Then, from inner product of matrices, we have $\text{Tr} U f_j f_j^* U^* E_{pq} = \text{Tr} U \tilde{f}_j \tilde{f}_j^* U^* E_{pq}$. Since this holds for all standard basis E_{pq} , We conclude that $U f_j f_j^* U^* = U \tilde{f}_j \tilde{f}_j^* U^*$ and thus $f_j f_j^* = \tilde{f}_j \tilde{f}_j^*$. Next, since $f_j f_j^*, \tilde{f}_j \tilde{f}_j^*$ are rank one projections, we have $\text{span } f_j = \text{span } \tilde{f}_j$. Then, $f_j = \lambda_j \tilde{f}_j$ and from $\|f_j\| = \|\tilde{f}_j\| = 1$, we have $|\lambda_j| = 1$. Now from construction, $D_1 = [|f_{ij}|^2]$ and $D_2 = [|\tilde{f}_{ij}|^2]$. Then, $D_1 = D_2$ $\square$

Lemma 2. *Given linearly independent matrices $\{K_i^* K_j\}_{ij}$ with $\sum_i K_i^* K_i = I$. Then, there exists an isometry $\mu = (\mu_{ij})_{ij}$ such that $\sum_i \mu_{ji} K_i = T_j$ if and only if $T_j \in \text{span } K_i$ and $\sum_j T_j^* T_j = I$*

Proof. For the forward implication, let μ be an isometry satisfying the conditions. Then, it is obvious that $T_j \in \text{span } K_i$. Since $\langle T_i, K_j \rangle = \mu_{ij}$, and $\mu^* \mu = I$, Then, we have $\sum_i T_i^* T_i = \sum_i (\sum_p \mu_{ip} K_p)^* (\sum_q \mu_{iq} K_q) = \sum_{pq} (\sum_i \overline{\mu_{ip}} \mu_{iq} K_p^* K_q) = \sum_{pq} \delta_{pq} K_p^* K_q = \sum_j K_j^* K_j = I$.

Conversely, we observe since $T_j \in \text{span } K_i$, let $T_j = \sum_i \mu_{ji} K_i$. Then, $\sum_j K_j^* K_j = I = \sum_i T_i^* T_i = \sum_i (\sum_p \mu_{ip} K_p)^* (\sum_q \mu_{iq} K_q) = \sum_{pq} (\mu^* \mu)_{pq} K_p^* K_q$. Then, it implies that $\sum_{pq} ((\mu^* \mu)_{pq} - \delta_{pq}) K_p^* K_q = 0$. Since $\{K_i^* K_j\}_{ij}$ is linearly independent, we have $(\mu^* \mu)_{pq} = \delta_{pq}$ and μ is an isometry. $\square$

3 Quick Recap

For now, we have defined $\mathbf{ICP}[A, B]$ be the set of unital completely maps satisfying $\Phi(A) = B$ and we defined $\mathbf{CS}[A, B]$ be the column stochastic matrix satisfying $b = aD$ where a, b are eigenvalues of A, B , respectively. $\mathbf{ICP}[A, B]$ is non-empty if and only if $\mathbf{CS}[A, B]$ is non-empty. Also, there is a $\tau : \mathbf{CS}[A, B] \rightarrow \mathbf{CP}[M_n, M_m, I]$ is injective. Moreover, we have $\text{Im } \tau \subset \text{ext}(\mathbf{ICP}[A, B])$

However, each τ depends on A, B , which makes our discussion hard. Our next question relies on, is there a universal τ or a τ that is independent to A, B .

4 New Ideas

At qubit case $\mathbf{M}_2$, we have Alberti-Uhlmann theorem, which states the followings are equivalent. Let A_1, A_2, B_1, B_2 be qubits. [3]

1. There is a TPCP map sending A_i to B_i for $i = 1, 2$.
2. $\|A_1 - tA_2\|_1 \geq \|B_1 - tB_2\|_1$ for all $t \geq 0$

This theorem only holds for qubit case. Now, we want to see a stronger condition that generalize (2) such that (1) holds for $\mathbf{M}_d$, $d \geq 3$.

Observe if (1) holds, we have $T \otimes I_d$ is positive, linear and trace preserving, and thus is a contraction with respect to trace norm(see *). Then,

$$\|B_1 \otimes I_d - B_2 \otimes T\|_1 = \|T \otimes I_d(A_1 \otimes I_d - A_2 \otimes T)\|_1 \leq \|A_1 \otimes I_d - A_2 \otimes T\|_1 \dots (3)$$

for all $T \geq 0$ and $T \in M_d$

Thus, we guess that (3) is the stronger condition generalize (2). To see this, suppose (3) holds and take $T = tI_d$, we have $\|B_1 \otimes I_d - B_2 \otimes T\|_1 = \|B_1 \otimes I_d - tB_2 \otimes I_d\|_1 = \|B_1 - tB_2\|_1$. Doing this operation for both side, we have (3) $\implies$ (2). Next, to see the equivalence in qubit case, observe that (3) $\implies$ (2) $\implies$ (1) $\implies$ (3). Thus, there equivalence is proved.

Another conditions we need to check is from [2], which states that the followings are equivalent. Let A, B be density matrices of $\mathbf{M}_d$, then

- i There is a unital TPCP map sending A to B for $i = 1, 2$.
- ii $B \prec A$

Note that (ii) states that A majorized B , which can be reformulated as $\|B - tI\|_1 \leq \|A - tI\|_1$, generalizing the Alberti-Uhlmann theorem when we set A_2, B_2 to be identities.

We will check our guessed condition satisfy this results. To see this, we first set $A_2 = B_2 = I_d$. Then, similarly to the previous section, we have $(3) \implies (ii)$. Thus, we conclude that $(3) \implies (ii) \implies (i) \implies (3)$

(*) We will show that positive, linear, TP map is trace norm contraction. Let T be such a map and given a matrix $A = A_+ - A_-$, where A_+, A_- are positive matrices. Then,

$$\|T(A)\|_1 \leq \|T(A_+)\|_1 + \|T(A_-)\|_1 = \text{Tr } T(A_+) + \text{Tr } T(A_-) = \text{Tr } A_+ + A_- = \text{Tr } |A| = \|A\|_1$$

References

- [1] Man-Duen Choi. Completely positive linear maps on complex matrices. *Linear Algebra and Its Applications*, 1975.
- [2] Chi-Kwong Li and Yiu-Tung Poon. Interpolation by completely positive maps. *Linear and Multilinear Algebra*, 2011.
- [3] et al Teiko Heinosaari. Extending quantum operations. *Journal of Mathematical Physics*, 2012.